

RQ Answers

omitted

2024-04-05

Data overview

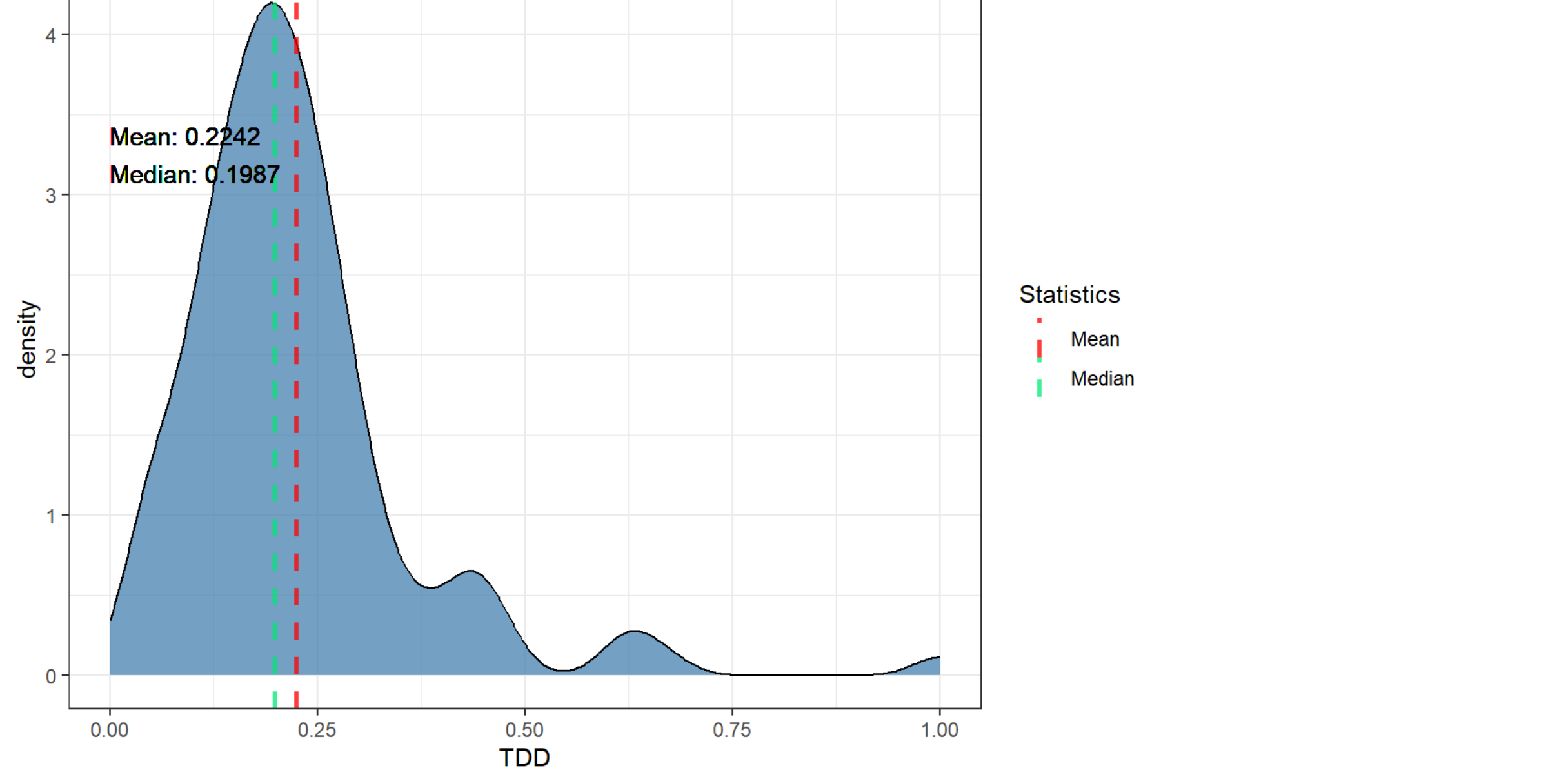
glimpse(all_data)				
## Rows: 1,779				
## Columns: 11				
## \$ repo	<chr> "httpcomponents-client", "httpcomponents-client", ...			
## \$ pr_number	<int> 118, 124, 124, 124, 133, 133, 133, 141, 141, 141, ...			
## \$ rule	<chr> "java:S3077", "java:S4165", "java:S1172", "java:S3...			
## \$ severity	<chr> "MINOR", "MAJOR", "MAJOR", "CRITICAL", "MAJOR", "M...			
## \$ file	<chr> "httpClient5/src/main/java/org/apache/hc/client5/h...			
## \$ type	<chr> "BUG", "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "...			
## \$ status	<chr> "OPEN", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPEN"...			
## \$ debt	<int> 20, 5, 5, 6, 5, 5, 5, 6, 5, 5, 1, 6, 10, 5, 5, 5, ...			
## \$ ncloc_affected_file	<int> 246, 107, 107, 107, 92, 92, 92, 350, 350, 350, 350...			
## \$ complexity	<int> 43, 20, 20, 20, 17, 17, 17, 94, 94, 94, 94, 63...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 110 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 110 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

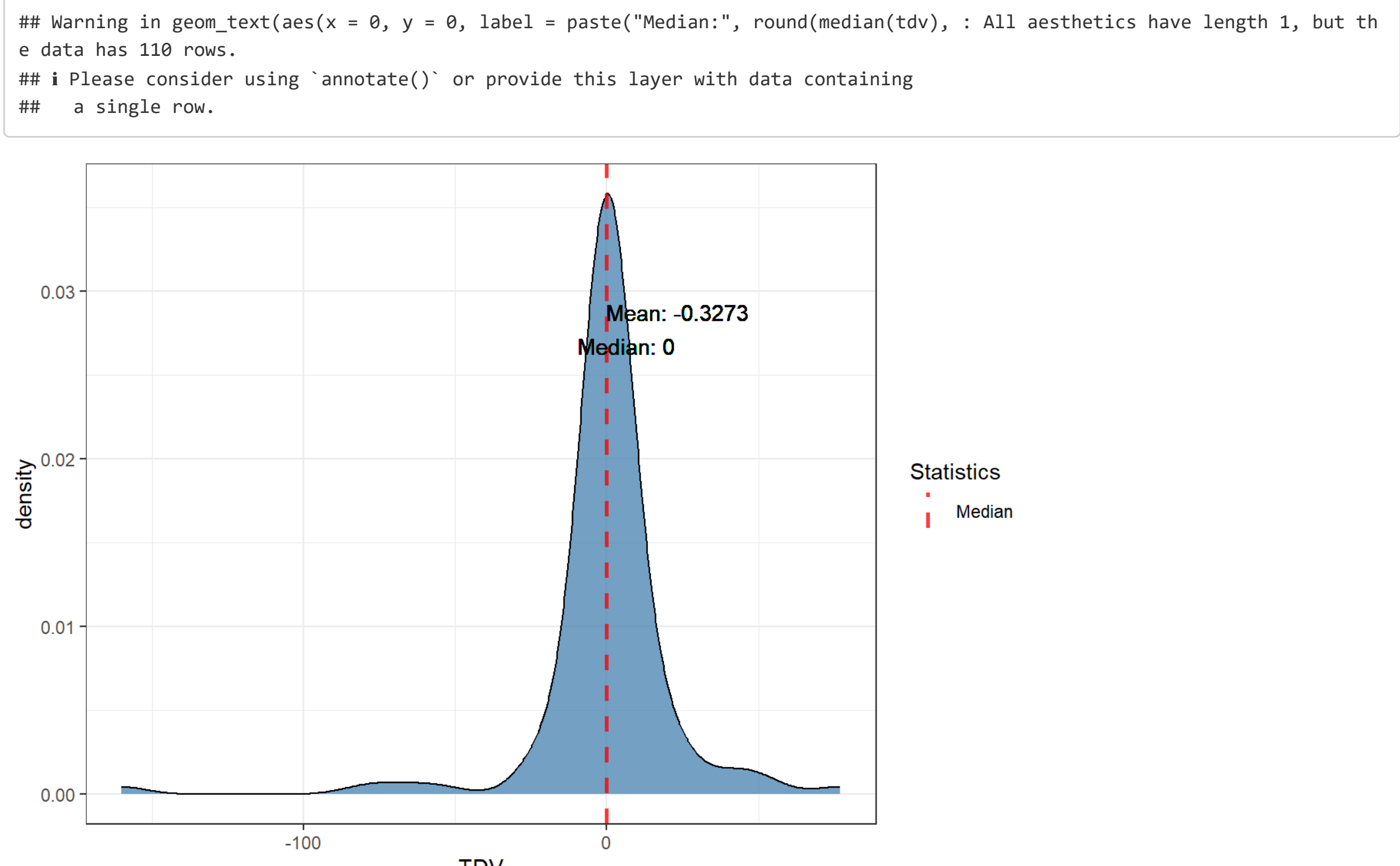


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 110 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 110 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero
1 httpcomponents-cl...	16.4	61.8	21.8

Mean of percentages of TDV (mean of repo percentages)

mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero
16.4	61.8	21.8

Percentages of pre-existing TDV by repo

repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_0
1 httpcomponents-client	75.7	24.3

Mean of percentages of pre-existing TDV (mean of repo percentages)

preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero
75.7	24.3

Percentages for new TDV by repo

repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero
1 httpcomponents-client	93.1	6.9

Mean of percentages of new TDV (mean of repo percentages)

mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero
93.1	6.9

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its $PositionScore_i$ will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the $PositionScore$, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

rank	httpcomponents-client
<int>	<chr>
1	1 java:S100 - 16
2	2 java:S3776 - 9
3	3 java:S1068 - 6
4	4 java:S1066 - 6
5	5 java:S1168 - 5
6	6 java:S1854 - 5
7	7 java:S1481 - 5
8	8 java:S107 - 4
9	9 java:S1124 - 3
10	10 java:S1192 - 3

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

rule	position_score
<chr>	<dbl>
1 java:S100	122
2 java:S3776	123
3 java:S1066	124
4 java:S1068	125
5 java:S1168	126
6 java:S1481	127
7 java:S1854	128
8 java:S107	129
9 java:S1124	130
10 java:S1192	131

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

rank	httpcomponents-client
<int>	<chr>
1	1 java:S3776 - 323
2	2 java:S1133 - 174
3	3 java:S1192 - 145
4	4 java:S1186 - 103
5	5 java:S1066 - 91
6	6 java:S107 - 76
7	7 java:S1124 - 73
8	8 java:S1117 - 71
9	9 java:S100 - 38
10	10 java:S3077 - 38

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

rule	position_score
<chr>	<dbl>
1 java:S3776	122
2 java:S1133	123
3 java:S1192	124
4 java:S1186	125
5 java:S1066	126
6 java:S107	127
7 java:S1124	128
8 java:S1117	129
9 java:S100	130
10 java:S3077	131

Outliers

Higher TDD

repo	pr_number	total_debt	total_ncloc	tdd
<chr>	<int>	<int>	<int>	<dbl>
1 httpcomponents-client	222	179	179	1
2 httpcomponents-client	212	244	366	0.667
3 httpcomponents-client	490	10	16	0.625
4 httpcomponents-client	504	245	396	0.619
5 httpcomponents-client	375	80	168	0.476
6 httpcomponents-client	370	991	2178	0.455
7 httpcomponents-client	541	621	1387	0.448
8 httpcomponents-client	356	115	259	0.444
9 httpcomponents-client	545	186	438	0.425
10 httpcomponents-client	480	115	273	0.421

Higher TDV

pr_number	repo	tdv
<int>	<chr>	<int>
1	543 httpcomponents-client	77
2	501 httpcomponents-client	51
3	421 httpcomponents-client	50
4	160 httpcomponents-client	40
5	208 httpcomponents-client	40
6	552 httpcomponents-client	28
7	542 httpcomponents-client	22
8	205 httpcomponents-client	20
9	506 httpcomponents-client	20
10	508 httpcomponents-client	20

Lower TDV

pr_number	repo	tdv
<int>	<chr>	<int>
1	370 httpcomponents-client	-160
2	202 httpcomponents-client	-80
3	483 httpcomponents-client	-68
4	547 httpcomponents-client	-55
5	537 httpcomponents-client	-25
6	496 httpcomponents-client	-21
7	345 httpcomponents-client	-20
8	377 httpcomponents-client	-20
9	409 httpcomponents-client	-20
10	312 httpcomponents-client	-15

Extra

Number of PRs with pre-existing TD

n
1 107

Number of java:S1192 issues that affect test code

n
1 0

Number of java:S1192 issues that affect not test code

n
1 148

Summary NCLOC

ncloc
Min.: 16.0
1st Qu.: 209.2
Median.: 361.0
Mean.: 879.4
3rd Qu.: 750.0
Max.: 10508.0